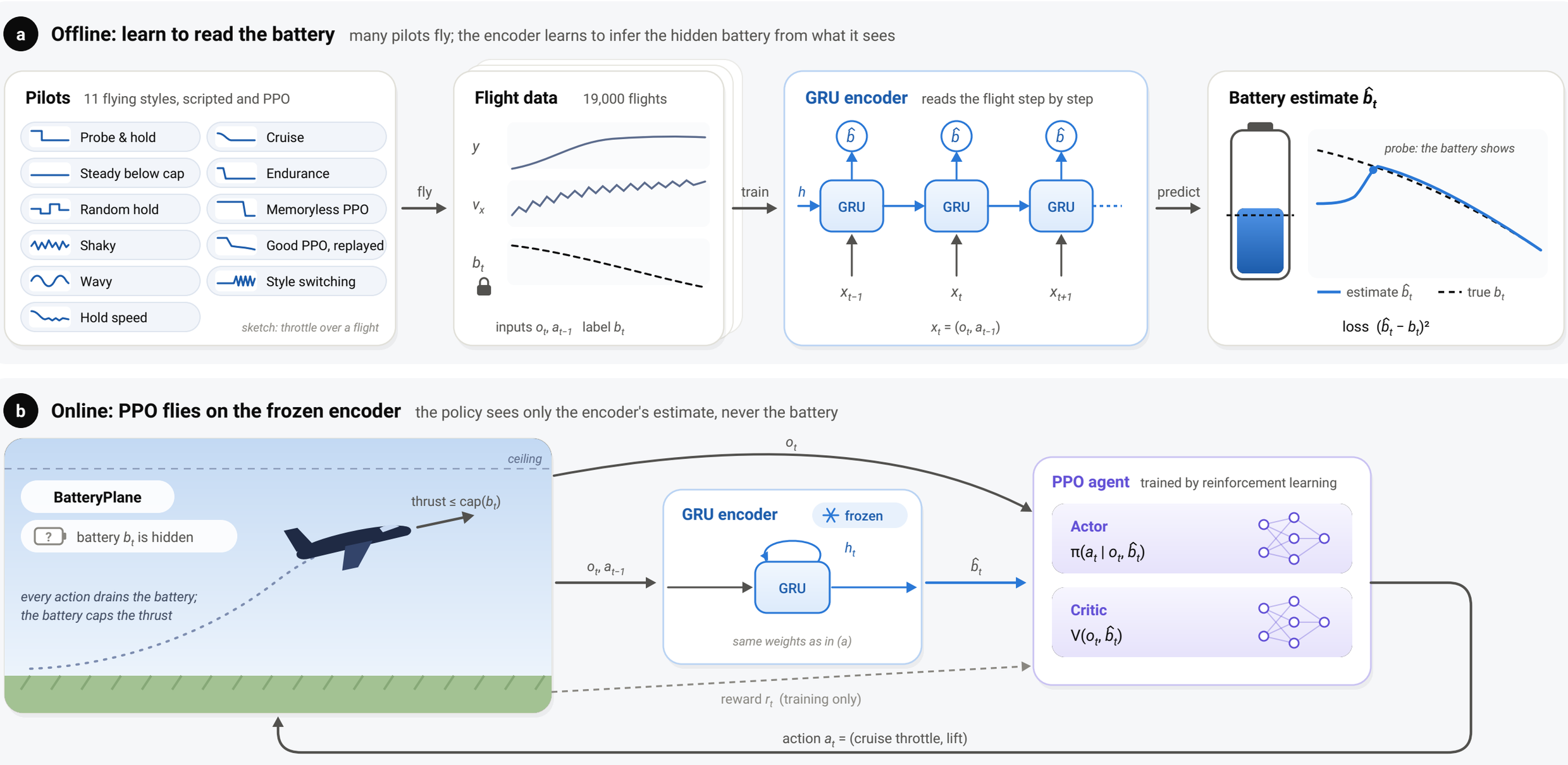

a
Offline: learn to read the battery
many pilots fly; the encoder learns to infer the hidden battery from what it sees
Pilots
11 flying styles, scripted and PPO
Probe & hold
Cruise
Steady below cap
Endurance
Random hold
Memoryless PPO
Shaky
Good PPO, replayed
Wavy
Style switching
Hold speed
sketch: throttle over a flight
fly
Flight data
19,000 flights
y
v_x
b_t
inputs o_t, a_{t-1} label b_t
train
GRU encoder
reads the flight step by step
$\hat{b}$
h
GRU
GRU
GRU
x_{t-1}
x_t
x_{t+1}
$x_t = (o_t, a_{t-1})$
predict
Battery estimate $\hat{b}_t$
probe: the battery shows
estimate $\hat{b}_t$
true b_t
loss $(\hat{b}_t - b_t)^2$
b
Online: PPO flies on the frozen encoder
the policy sees only the encoder's estimate, never the battery
ceiling
BatteryPlane
battery b_t is hidden
thrust ≤ cap(b_t)
every action drains the battery;
the battery caps the thrust
o_t
o_t, a_{t-1}
GRU encoder
frozen
h_t
GRU
same weights as in (a)
$\hat{b}_t$
PPO agent
trained by reinforcement learning
Actor
$\pi(a_t \mid o_t, \hat{b}_t)$
Critic
$V(o_t, \hat{b}_t)$
reward r_t (training only)
action a_t = (cruise throttle, lift)